Codefest 6

CMAN — Sneak Paths in a Resistive Crossbar

(a) In order for current to reach ground on col 0 it must only pass through the top 1 kΩ resistor (ideally) and hence by Ohm's law:

I_{col0} = 1V / 1kΩ = **1 mA** output.

(b) Since row 0 is kept at 1V, the current can also travel via the sneak path which jumps to col 1 via the 2 kΩ resistor, then on to row 1 via the 1 kΩ resistor, and then to ground via the 2 kΩ resistor. Hence:

$I_{sneak} = V / (R_2 + R_3 + R_4)$ = 1V / 5 kΩ = **0.2 mA**

$V_{row1} = I_{sneak} \times R_3$ = 2/5 mV = **0.4 mV**

As for V_{col1}, start with V_{row0} at 1V and dropping through the 2 kΩ resistor gives:

V_{col1} = (1 – 2/5) V = 3/5 V = **0.6 V**

(c) $I_{col0} = I_{ideal} + I_{sneak}$ = (1 + 0.2) mA = **1.2 mA**

(d) Ideally, the current in each column should be only a function of the values of the resistors that lie in that column. This is what makes this model work as a matrix-vector multiply algorithm. With the sneak current this is no longer strictly true.